

# ANIL PRADHAN

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AI & Backend Engineer specializing in autonomous multi-agent architectures, computer vision pipelines, and resilient backend systems with Python, FastAPI, LangGraph, and PyTorch.

Dell FutureMinds 2nd Runner-Up (Top 6 / 8,000+) · 4× AI Hackathon Builder (Dell, MSFT, AWS, Google)

## EDUCATION

### Institute of Technical Education & Research (ITER), SOA University

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 – 2028

CGPA: 8.93

### Nalanda Vidya Mandir Higher Secondary School, Berhampur, Odisha

Class XII (Higher Secondary) — Physics, Chemistry, Mathematics with IT

2022 – 2024

Score: 85%

## TECHNICAL PROJECTS

### VisionForge AI

Aug 2026 – Present

[GitHub](#) — AI/ML & Pipeline Engineer | Team: Disha & Anil

- Built an autonomous inspection pipeline using **LangGraph state machines** to run OCR, seal-matching, and YOLO11n detection in parallel, with **Groq and Gemini Vision** models giving the final pass/fail verdict in under 5s.
- Trained and fine-tuned a custom **YOLO11n object-detection model on 4,448 images**, reaching **98.4% mAP@50** for identifying and locating hardware parts.
- Designed a custom evidence-fusion scoring method (**Anomaly Max-Pooling**) combined with EasyOCR text verification and Gemini embeddings + FAISS similarity search for reference matching — so no defect gets diluted or missed — validated with **203 automated tests**.

**Tech:** Python · FastAPI · LangGraph · PyTorch · YOLO11n · OpenCV · CLIP · FAISS · Groq · Gemini

### TaskPilot AI

Jul 2026 – Present

[Live Demo](#) | [GitHub](#) — Integration Lead | Team IdeaForge

- Built a **6-stage multi-agent pipeline** (Agent 0 Orchestrator) in Python & FastAPI to collect, extract, clean, and schedule tasks from 5 tools in **~20 seconds**.
- Developed **smart deduplication and a 7-factor priority engine** (severity, deadline, outage risk, customer impact) with a 7-point quality checker to rank tasks with clear explanations.
- Implemented a **reliable LLM client with circuit breaker** (Groq + local fallback rules) and an AI chat assistant for **instant P1 bug injection** and calendar-aware daily scheduling.

**Tech:** FastAPI · React · SQLite · Groq/NVIDIA NIM

### AI Career Mentor

Feb 2026 – Aug 2026

[Live Demo](#) | [GitHub](#) — Solo Developer

- Built a multi-agent AI system using LangGraph to orchestrate 4 parallel agents (Resume, Market, Roadmap, LinkedIn) with shared state, reducing analysis latency by ~60%.
- Developed a real-time mock interview engine with a 7-phase FSM over WebSockets, featuring a live code editor and resume-personalized questions.
- Designed a fault-tolerant backend (FastAPI, PostgreSQL, Redis) with RAG-powered roadmap generation via ChromaDB, multi-provider LLM failover, and a 113-test CI/CD pipeline.

**Tech:** FastAPI · LangGraph · Next.js · ChromaDB · Redis · Docker · ONNX

## TRAINING

### Agentic AI Training Program — CogniForce.AI & ITER, SOA University

2026 – Present

- Hands-on training in Agentic AI, LLMs, prompt engineering, RAG, tool calling, MCP, and multi-agent orchestration.
- Built and experimented with agents using OpenClaw, Multica, AnythingLLM, OpenRouter, Claude Code, OpenCode, and MCP servers, including agent squads and tool integrations.

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL

**Backend & APIs:** FastAPI, REST APIs, SQLAlchemy, Pydantic v2, JWT/RBAC, SSE, Alembic

**Gen AI & Agentic AI:** LangGraph, Multi-Agent Systems, RAG, Semantic Search, Prompt Engineering, LLM Integration

**Computer Vision & Embeddings:** PyTorch, OpenCV, YOLO11n, CLIP, FAISS, OCR, Anomaly Detection (SSIM)

**Databases:** PostgreSQL, Redis, ChromaDB, SQLite

**DevOps & Tools:** Docker, GitHub Actions (CI/CD), Git/GitHub, VS Code, Claude Code, Antigravity, Linux/Bash

**Core CS:** DSA, OOP, OS, DBMS, CN

**Soft Skills:** Problem-Solving, Technical Communication, Teamwork, Time Management

## ACHIEVEMENTS & CERTIFICATIONS

- Dell FutureMinds AI Hackathon 2026 — 2nd Runner-Up (3rd Place):** Advanced through a 4-stage national selection (8,000+ participants → 370 → Top 10 → Top 6 Grand Finalists); presented final solution to Dell jury and senior leaders at Dell Bengaluru.
- FlowFinance × Techfest IIT Bombay — Round 2 Qualifier:** Designed a GTM strategy deck (platform analysis, customer personas, acquisition/retention framework) for FlowFinance (AI SMB fintech platform) with real customer outreach.
- 4× AI Hackathon Builder:** Built and deployed AI solutions across 4 hackathons, including Microsoft AI Dev Days, Amazon Nova Challenge, and Google Solution Challenge.
- 100+ Day DSA Streak:** Maintained a 100+ day coding streak on GeeksforGeeks' 160 Days of Code Challenge, solving DSA problems daily — recognized with GFG consistency reward.
- Certifications:** IBM AI Fundamentals · Cisco Python Essentials 2 · Wells Fargo SWE (Forage) · HackerRank (Problem Solving, SQL, Java, Python).